JP1: BRIDGE ONE SIDE ONLY. JP2: REGULATOR OR A105 POWER, NEVER BOTH.

SYS_3V3
GPIO4_PWM
GPIO6
GPIO8
GND
UART_RX_GPIO20
GPIO18
GPIO3
GPIO1
GND
U1
ESP32-C3-WROOM-02U-N4
EN
GPIO5
GPIO7
GPIO9_BOOT
GPIO10
UART_TX_GPIO21
GPIO19
GPIO2
GPIO0

GND
REG_IN
BUCK_FB
BUCK_BOOT
GND
U2
LMR36510FADDAR
REG_IN
BUCK_VCC
BUCK_SW

DC_RAW
GND
J1
DC 1024 compatible 5.5/2.1mm
GND

GND
USB_5V
GND
USB_5V
GND
J2
USB-C POWER ONLY
GND
CC1
USB_5V
GND
CC2
USB_5V

DC_RAW
LED OUT
J3
LED_NEG

GND
EN
UART_TX_GPIO21
M5STACK A105
J4
GPIO9_BOOT
UART_RX_GPIO20
PROG_3V3

GND
GPIO0
GPIO2
GPIO4_PWM
GPIO6
GPIO8
DIP LEFT
JH1
SYS_3V3
GPIO1
GPIO3
GPIO5
GPIO7
GND

GND
GPIO18
UART_RX_GPIO20
GPIO10
DIP RIGHT
JH2
GPIO19
UART_TX_GPIO21
SYS_3V3
GND

USB_5V
DC_RAW
INPUT USB|DC
JP1
REG_IN

REG_3V3
PROG_3V3
3V3 REG|A105
JP2
SYS_3V3

MOS_GATE
GND
RJP020N06T100
Q1
LED_NEG

BUCK_VCC
ZETC00470S-220MT-LF 22uH 2A Isat 4A
REG_3V3

REG_3V3
57.6k 1%
RFBT
BUCK_FB

BUCK_FB
24.9k 1%
RFBB
GND

SYS_3V3
10k
REN
EN

SYS_3V3
10k
RBOOT
GPIO9_BOOT

GND
5.1k 1%
RCC1
CC1

GND
5.1k 1%
RCC2
CC2

GPIO4_PWM
100R
RG
MOS_GATE

MOS_GATE
100k
RPD
GND

REG_IN
2.2uF 50V
CIN1
GND

REG_IN
220nF 100V
CIN2
GND

BUCK_BOOT
100nF 25V
CBOOT
BUCK_SW

BUCK_VCC
1uF 10V
CVCC
GND

REG_3V3
22uF 10V
COUT1
GND

REG_3V3
22uF 10V
COUT2
GND

SYS_3V3
10uF 10V
C3V3
GND

SYS_3V3
100nF
CDEC
GND

EN
1uF 10V
CEN
GND

Power-only USB-C; selectable USB/DC input; selectable regulator/A105 3V3

Sheet: /
File: esp32-c3-compact-dip-led.kicad_sch

Title: ESP32-C3 Compact DIP + PWM

Size: A3	Date:	Rev:
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